SUPPLEMENT

Supplementary Table 1. Specific Targets for Molecular Analysis

Study	Molecular Analysis						
	EGFR-TKI (%)	EGFR Positivity (%)	T790M Positivity V (%)	ALK Rearrangement (%)	KRAS (%)	RET (%)	ROS1 (%)
Solomon et al. 2010 [1]	ND	5/16 (31)	NS	NS	NS	NS	NS
Zhuang et al. 2011 [2]	ND	23/43 (53)	NS	NS	NS	NS	NS
Yoon et al. 2012 [3]	54/94 (57)	20/41 (49)	5/41 (12)	11/40 (28)	ND	ND	ND
Hsiao et al. 2013 [4]	ND	79/132 (60)	NS	NS	NS	NS	NS
Tam et al. 2013 [5]	ND	NS	NS	NS	NS	NS	NS
Schneider et al. 2015 [6]	ND	NS	NS	NS	NS	NS	NS
Florentine et al. 2015 [7]	ND	NS	NS	NS	NS	NS	NS
Nosaki et al. 2016 [8]	395/395 (100)	283/296 (96)	NS	NS	NS	NS	NS
Tokaca et al. 2018 [9]	NS	26 (14/52)	ND	39/182 (21)*	31/182 (17)*	2/182 (1)*	3/182 (2)*
Hata et al. 2017 [10]	NS	NS	ND	ND	ND	ND	ND
Tian et al. 2017 [11]	ND	72/168 (43)	5/66 (8)	NS	NS	NS	NS
Matsumoto et al. 2018 [12]	NS	NS	5/10 (50)	ND	ND	ND	ND
Komiya et al. 2018 [13]	22/22 (100)	5/5 (100)	2/5 (40)	ND	ND	ND	ND
Seto et al. 2018 [14]	152/236 (64)	100 (12/12)	NS	ND	ND	ND	ND
Nam et al. 2019 [15]	156/199 (78)	117/179 (65)	59/179 (33)	ND	ND	ND	ND
Gill et al. 2018 [16]	NS	NS	NS	NS	NS	ND	ND
Kim et al. 2018 [17]	NS	NS	50/81 (62)	ND	ND	ND	ND
Fintelmann et al. 2019 [18]	NS	NS	NS	ND	ND	ND	ND
Hong et al. 2019 [19]	98/112 (88)	NS	12/18 (67)	ND	ND	ND	ND
Lee et al. 2019 [20]	338/338 (100)	NS	62/129 (48)	NS	NS	NS	NS
Beck et al. 2019 [21]	ND	31/96 (32)	NS	4/86 (5)	2/34 (6)	1/11 (9)	0/11 (0)

*The total number of genes with successful molecular analysis was 182 in this study. ALK = anaplastic lymphoma kinase, EGFR = epidermal growth factor receptor, KRAS = Kirsten rat sarcoma viral oncogene homolog, ND = not done, NS = not specified, RET = rearranged during transfection, ROS1 = c-ros oncogene 1

REFERENCES

1. Solomon SB, Zakowski MF, Pao W, Thornton RH, Ladanyi M, Kris MG, et al. Core needle lung biopsy specimens: adequacy for EGFR and KRAS mutational analysis. *AJR Am J Roentgenol* 2010;194:266-269
2. Zhuang YP, Wang HY, Shi MQ, Zhang J, Feng Y. Use of CTguided fine needle aspiration biopsy in epidermal growth factor receptor mutation analysis in patients with advanced lung cancer. *Acta Radiol* 2011;52:1083-1087
3. Yoon HJ, Lee HY, Lee KS, Choi YL, Ahn MJ, Park K, et al. Repeat biopsy for mutational analysis of non-small cell lung cancers resistant to previous chemotherapy: adequacy and complications. *Radiology* 2012;265:939-948
4. Hsiao SH, Chung CL, Lee CM, Chen WY, Chou YT, Wu ZH, et al. Suitability of computed tomography-guided biopsy specimens for subtyping and genotyping of non-small-cell lung cancer. *Clin Lung Cancer* 2013;14:719-725
5. Tam AL, Kim ES, Lee JJ, Ensor JE, Hicks ME, Tang X, et al. Feasibility of image-guided transthoracic core-needle biopsy in the BATTLE lung trial. *J Thorac Oncol* 2013;8:436-442
6. Schneider F, Smith MA, Lane MC, Pantanowitz L, Dacic S, Ohori NP. Adequacy of core needle biopsy specimens and fine-needle aspirates for molecular testing of lung adenocarcinomas. *Am J Clin Pathol* 2015;143:193-200; quiz 306
7. Florentine BD, Helton RJ, Mitchell MM, Schmidt KE, Kozlov DB. Accuracy and adequacy of computed tomography-guided lung biopsies: experience from a community hospital. *J Am Osteopath Assoc* 2015;115:592-603
8. Nosaki K, Satouchi M, Kurata T, Yoshida T, Okamoto I, Katakami N, et al. Re-biopsy status among non-small cell lung cancer patients in Japan: a retrospective study. *Lung Cancer* 2016;101:1-8
9. Tokaca N, Barth S, O'Brien M, Bhosle J, Fotiadis N, Wotherspoon A, et al. Molecular adequacy of image-guided rebiopsies for molecular retesting in advanced non-small cell lung cancer: a single-center experience. *J Thorac Oncol* 2018;13:63-72
10. Hata A, Katakami N, Nanjo S, Okuda C, Kaji R, Imai Y. Rebiopsy of histological samples in pretreated non-small cell lung cancer: comparison among rebiopsy procedures. *In Vivo* 2017;31:475-479
11. Tian P, Wang Y, Li L, Zhou Y, Luo W, Li W. CT-guided transthoracic core needle biopsy for small pulmonary lesions: diagnostic performance and adequacy for molecular testing. *J Thorac Dis* 2017;9:333-343
12. Matsumoto T, Hasebe T, Baba Y, Chosa K, Kondo S, Yamada S, et al. Feasibility and safety of CT-guided intrathoracic and bone re-biopsy for non-small cell lung cancer. *Anticancer Res* 2018;38:3587-3592
13. Komiya K, Nakashima C, Nakamura T, Hirakawa H, Abe T, Ogusu S, et al. Current status and problems of T790M detection, a molecular biomarker of acquired resistance to EGFR tyrosine kinase inhibitors, with liquid biopsy and rebiopsy. *Anticancer Res* 2018;38:3559-3566
14. Seto T, Nogami N, Yamamoto N, Atagi S, Tashiro N, Yoshimura Y, et al. Real-world EGFR T790M testing in advanced nonsmall-cell lung cancer: a prospective observational study in Japan. *Oncol Ther* 2018;6:203-215
15. Nam BD, Kim TJ, Park K, Ahn MJ, Choi YL, Chung MJ, et al. Transthoracic rebiopsy for mutation analysis in lung adenocarcinoma: outcomes and risk factors for the acquisition of nondiagnostic specimens in 199 patients. *Clin Lung Cancer* 2019;20:e309-e316
16. Gill RR, Murphy DJ, Kravets S, Sholl LM, Janne PA, Johnson BE. Success of genomic profiling of non-small cell lung cancer biopsies obtained by trans-thoracic percutaneous needle biopsy. *J Surg Oncol* 2018;118:1170-1177
17. Kim H, Chae KJ, Yoon SH, Kim M, Keam B, Kim TM, et al. Repeat biopsy of patients with acquired resistance to EGFR TKIs: implications of biopsy-related factors on T790M mutation detection. *Eur Radiol* 2018;28:861-868
18. Fintelmann FJ, Troschel FM, Kuklinski MW, McDermott S, Petranovic M, Digumarthy SR, et al. Safety and success of repeat lung needle biopsies in patients with epidermal growth factor receptor-mutant lung cancer. *Oncologist* 2019;24:1570-1576
19. Hong MH, Kim HR, Ahn BC, Heo SJ, Kim JH, Cho BC. Realworld analysis of the efficacy of Rebiopsy and EGFR mutation test of tissue and plasma samples in drug-resistant non-small cell lung cancer. *Yonsei Med J* 2019;60:525-534
20. Lee K, Kim Y, Jung HA, Lee SH, Ahn JS, Ahn MJ, et al. Repeat biopsy procedures and T790M rates after afatinib, gefitinib, or erlotinib therapy in patients with lung cancer. *Lung Cancer* 2019;130:87-92
21. Beck KS, Kim TJ, Lee KY, Kim YK, Kang JH, Han DH. CTguided coaxial biopsy of malignant lung lesions: are cores from 20-gauge needle adequate for histologic diagnosis and molecular analysis? *J Thorac Dis* 2019;11:753-765